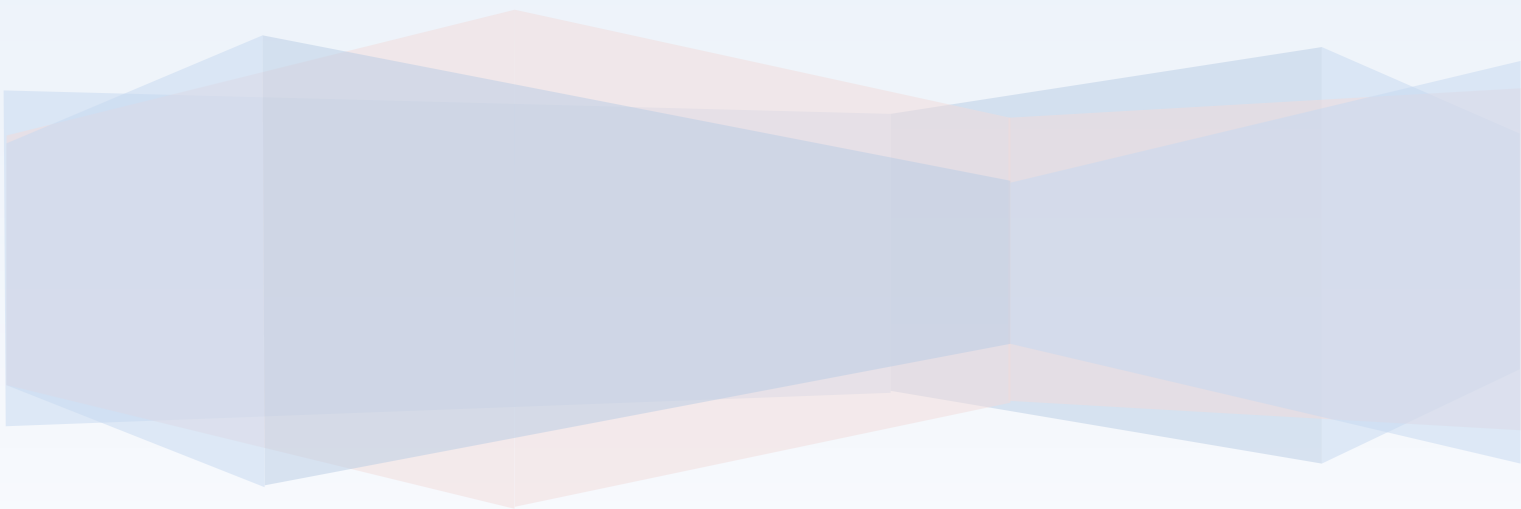


COS30043 – Interface Design and Development

Learning Summary Report

Tran Thanh Minh (103809048)



Self-Assessment Details

The following checklists provide an overview of my self-assessment for this unit.

	Pass (D)	Credit (C)	Distinction (D)	High Distinction (HD)
Self-Assessment (please tick)		✓		

Self-assessment Statement

	Included (please tick)
Learning Summary Report	✓
Use of Bootstrap that demonstrate coverage of core concepts	✓
Use of VueJS that demonstrate coverage of core concepts	✓

Minimum Pass Checklist

	Included (please tick)
Progress on Credit Tasks	✓
All Pass Tasks signed off	✓

Minimum Credit Checklist, in addition to Pass Checklist

	Included (please tick)
Credit and Pass Tasks done, and Progress on Distinction Tasks.	
Custom program meets Distinction criteria	
Design report with screenshots for custom program	

Minimum Distinction Checklist, in addition to Credit Checklist

	Included (please tick)
Research report, and associated pieces	
Custom project meets HD requirements	

Minimum High Distinction Checklist, in addition to Distinction Checklist

Declaration

I declare that this portfolio is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person.

Signature: _____ Tran Thanh Minh

Introduction

This report summarises what I learnt in COS30043 Interface Design and Development. It includes a self-assessment against the criteria described in the unit outline, a justification of the pieces included, details of the coverage of the unit's intended learning outcomes, and a reflection on my learning.

Overview of Pieces Included

This section outlines the pieces that I have included in my portfolio...

1. 1.1P: Submit old webproject have done in HTML/CSS/JS
2. 1.2P: 'Hello World' using HTML
3. 1.3P: Check the accessibility using the validator for HTML/CSS
4. 1.4P: Check the accessibility of table using the validator for HTML/CSS
5. 2.1P: Using Bootstrap and answer all questions
6. 2.2P: Design calculator with Bootstrap
7. 2.3P: Create Bootstrap template
8. 3.1P: Create string test app with VueJS
9. 3.2P: Create Lookup app with VueJS
10. 3.3C: Create a compute BMI app with VueJS
11. 3.4C: Cloud service Registration app with VueJS
12. 4.1P: Creating Guessing App with VueJS
13. 5.2P: Create a menu list app with VueJS
14. 5.3C: Create a app with router with VueJS
15. 6.1C: Create a registration with VueJS
16. 7.1P: Fetch Data from API using VueJS
17. 7.2P: Read data from JSON file using VueJS
18. 8.1P: Create pagination in VueJS
19. 8.2C: Read data from JSON file and apply pagination with VueJS
20. 9.1P: Create a single page application with VueJS with router
21. 9.2C: Create a single page application with VueJS include router, backend, database.
22. 10.1P: Create a single page application with Vue-CLI
23. 11.1P: Learning summary report

Coverage of the Intended Learning Outcomes

This section outlines how the pieces I have included demonstrate the depth of my understanding in relation to each of the unit's intended learning outcomes.

ILO 1: Apply Design

Apply fundamental design concepts and standards to the development of user interfaces

The following pieces demonstrate my ability in relation to this ILO:

- **2.2P:** Design calculator with Bootstrap: This project showcases my ability to create a functional and visually appealing user interface using Bootstrap components. I effectively applied grid layout, typography, and spacing to ensure a responsive and user-friendly calculator design.
- **2.3P:** Create Bootstrap template: This project highlights my understanding of design principles and Bootstrap's component library. I developed a visually coherent and responsive website template, demonstrating my ability to create a harmonious user experience across different screen sizes.
- **3.2P:** Create Lookup app with VueJS: This project demonstrates my ability to combine design principles with VueJS development. I employed Bootstrap's grid system to create a well-structured and responsive layout for the lookup app, enhancing user interaction and navigation.
- **3.4C:** Cloud service Registration app with VueJS: This project further solidifies my design skills within a VueJS framework. I utilized Bootstrap's grid system to create a

visually pleasing and responsive registration form, ensuring a positive user experience.

- **5.3C:** Create a app with router with VueJS: This project showcases my ability to apply design considerations to a single-page application (SPA) using Vue Router. I integrated Bootstrap components to create a consistent and intuitive navigation experience for users.
- **8.1C:** Create pagination in VueJS: This project showcases my ability to design interactive components. I used Vuejs-paginate-next to create a visually appealing and functional pagination component, enhancing user experience when navigating through large datasets.
- **9.2C:** Create a single page application with VueJS include router, backend, database: This project demonstrates my ability to design a comprehensive SPA with CRUD functionalities. I integrated Bootstrap components to create a visually appealing and user-friendly interface, considering user needs and preferences throughout the application.

ILO 2: Use Frameworks

Use contemporary frameworks to create dynamic user interfaces.

- **3.1P:** Use control directives to change messages based on input values for String Test web app: This project showcases my ability to utilize Vue.js directives to create dynamic and interactive user interfaces. By employing control directives, I effectively modified the displayed messages in response to user input, enhancing user engagement.
- **3.2C:** Use the loop directive to create a table from an array of data objects and use the filter directive to filter and search based on user input: This project demonstrates my proficiency in leveraging Vue.js directives for data manipulation and presentation. The loop directive enabled the dynamic generation of table rows from data objects, while the filter directive facilitated efficient search and filtering capabilities, providing a responsive user experience.
- **3.3C:** Use expression directive to calculate BMI from user input and condition directive to display message based on BMI value: This project highlights my understanding of Vue.js directives for data binding and conditional rendering. By utilizing expression and condition directives, I created a dynamic BMI calculator that displayed calculated results and corresponding messages, enhancing user interaction and information clarity.
- **3.4C:** Displays many different types of inputs depending on the user's selected information: This project further demonstrates my ability to create dynamic user interfaces using Vue.js directives. By dynamically rendering different input types based on user selections, I provided a flexible and user-friendly experience, adapting to user needs and preferences.
- **5.1P:** Use Vuejs component to handle adding and removing status of status posting app: This project showcases my ability to create reusable and maintainable components using Vue.js. By encapsulating the logic for adding and removing status updates within a component, I improved code organization and promoted component-based development.
- **5.2P:** Use prop to pass data from parent component to child component: This project demonstrates my understanding of component communication in Vue.js. By effectively utilizing props to pass data between parent and child components, I created a well-structured and efficient component hierarchy.
- **5.3C:** Implement router to navigate and display detailed information of each unit for the look-up unit app: This project highlights my ability to create single-page applications (SPAs) using Vue Router. By implementing navigation and dynamic content rendering, I enhanced user experience by providing seamless transitions and detailed information within a single page.
- **7.2P:** Using jQuery to retrieve data from the outside and display that data on the internal user interface: This project showcases my ability to integrate external data into a user interface using jQuery. By fetching data and dynamically updating the UI, I created a more interactive and informative user experience.

- **8.1P:** Implement vue-paginate-next to paginate student marks: This project highlights my ability to enhance user experience through pagination. By incorporating vue-paginate-next, I improved data presentation by allowing users to navigate through large datasets efficiently.
- **9.2C:** Create single page application with CRUD functionalities using components, vue-router, vue-paginate-next, and backend: This project demonstrates my comprehensive understanding of Vue.js framework and its capabilities for building complex SPAs. By combining components, routing, pagination, and backend integration, I created a robust and user-friendly application with CRUD operations.
- **10.1P:** Create single page application with component and router using vue-cli: This project showcases my ability to leverage Vue CLI for rapid SPA development. By utilizing components and routing, I efficiently created a well-structured and maintainable application.

ILO 3: Develop User Interfaces

Design and develop user interfaces optimised for a range of devices and platforms.

- **2.1P:** The computer application is built using Bootstrap, capable of adapting to all different screen sizes: This project showcases my ability to create responsive user interfaces using Bootstrap. By utilizing its row-col system and responsive design principles, I ensured that the application displayed correctly and functioned optimally across various devices and screen sizes.
- **2.3P:** Design scalable business website templates: This project demonstrates my ability to create flexible and adaptable website templates. By considering different screen sizes and device capabilities, I designed templates that could be easily scaled and customized for various platforms.

ILO 4: Evaluate User Interfaces

Evaluate user interfaces with respect to usability and accessibility using appropriate techniques, and propose improvements.

- **2.3P:** Design and implement interfaces, clearly identify the main parts of the website and the functions of each part. Focus on creating intuitiveness for the interface and ease of use for users: This project showcases my ability to evaluate user interfaces based on usability principles. By clearly defining website sections and their functions, I aimed to create an intuitive and user-friendly experience.

Reflection

The most important things I learnt:

During the Interface Design and Development course, I gained a comprehensive understanding of the fundamental principles and practices essential for creating effective and visually appealing user interfaces. A key takeaway was the significance of user-centric design, where understanding user needs and behaviors is paramount in crafting intuitive and engaging experiences. Additionally, I acquired proficiency in applying design concepts such as layout, color theory, and typography to enhance both aesthetic appeal and usability. Moreover, the course emphasized the importance of responsive design, enabling the creation of interfaces that adapt seamlessly to various devices and screen sizes.

The things that helped me most were:

Several factors contributed to my successful learning journey in this course. Firstly, the practical application of theoretical knowledge through hands-on projects was invaluable. By building real-world interfaces, I gained a deeper appreciation for the challenges and intricacies involved in the design process. Secondly, the availability of comprehensive resources and tools, such as design software and coding frameworks, facilitated experimentation and rapid prototyping. Lastly, the collaborative learning environment fostered by the course encouraged knowledge sharing and peer feedback, enriching my understanding of different design perspectives.

I found the following topics particularly challenging:

While the course provided a solid foundation in interface design, I encountered some challenges in balancing aesthetic appeal with optimal usability. Striking the right equilibrium between visually pleasing designs and intuitive user interactions required careful consideration and iterative testing. Additionally, staying updated with the latest design trends and technologies while maintaining a focus on core principles proved to be a demanding task.

I found the following topics particularly interesting:

I was particularly captivated by the exploration of user experience (UX) design principles and their impact on user satisfaction. Understanding how to create intuitive and enjoyable interactions through careful information architecture and interaction design was fascinating. Moreover, the integration of emerging technologies, such as virtual reality and augmented reality, into interface design opened up exciting possibilities for innovative user experiences.

I feel I learnt these topics, concepts, and/or tools really well:

I have developed a strong foundation in fundamental design principles, including layout, color theory, typography, and responsive design. My ability to create visually appealing and user-friendly interfaces has significantly improved. Furthermore, I have gained proficiency in utilizing design tools and software to efficiently create and iterate on design concepts. My understanding of user-centered design methodologies has also deepened, enabling me to empathize with users and design solutions that meet their needs effectively.

I still need to work on the following areas:

While I have made significant progress, I recognize the need for further development in certain areas. I would like to deepen my knowledge of advanced usability testing techniques to gain more insights into user behaviors and preferences. Additionally, exploring emerging interaction design paradigms and staying updated with the latest design trends will be essential for staying competitive in the field.

My progress in this unit was ...:

Overall, my progress in the Interface Design and Development unit has been positive and rewarding. I have acquired a solid foundation in design principles and practical skills, which have equipped me with the confidence to tackle complex interface design challenges. I am eager to apply my newfound knowledge to real-world projects and continue expanding my design expertise.

This unit will help me in the future:

The skills and knowledge gained from this unit are directly applicable to various fields, including web development, user experience design, and human-computer interaction. I believe this foundation will be invaluable in my future endeavors, enabling me to create user-centric products and services that exceed user expectations.

If I did this unit again I would do the following things differently:

If I had the opportunity to take this course again, I would allocate more time to exploring advanced topics such as information architecture and interaction design patterns. Additionally, I would seek out opportunities to collaborate with other designers and developers to gain diverse perspectives and enhance my problem-solving abilities.

Other...:

I believe that continuous learning and staying updated with the latest design trends are essential for success in this field. I plan to actively participate in design communities and attend industry conferences to expand my knowledge and network with other professionals.

By reflecting on my learning journey, I have identified areas of strength and areas for improvement. I am excited to apply my newfound knowledge and skills to future projects and contribute to the creation of exceptional user experiences.